

Conferences > 2021 IEEE International Confe... 

Qualitative Evaluation of Face Embeddings Extracted From well-known Face Recognition Models

Publisher: IEEE [Cite This](#)  PDF

Rassul Tolegenov ; Kairat Bostanbekov ; Daniyar Nurseitov ; Kuanysh Slyamkhan All Authors ...

1
Cites in
Paper

177
Full
Text Views

    

Alerts

Manage Content Alerts
Add to Citation Alerts

Abstract

Document Sections

I. INTRODUCTION

» DEEP LEARNING BASED FACE RECOGNITION MODEL

» EXPERIMENTS AND EVALUATION

» DISCUSSION AND CONCLUSION

Authors

Figures

References

Citations

Keywords

Metrics

More Like This

 Download PDF

Abstract:This paper demonstrates a qualitative evaluation/comparison of face embeddings extracted from deep learning models, such as VGG-Face, Dlib, and OpenFace, on a face discri... [View more](#)

► **Metadata**

Abstract:

This paper demonstrates a qualitative evaluation/comparison of face embeddings extracted from deep learning models, such as VGG-Face, Dlib, and OpenFace, on a face discrimination task. While conducting experiments, each of linear SVM (support vector machine) classifier, Euclidean distance, and Cosine distance algorithms was utilized to compare/analyze face vectors (embeddings) extracted from those 3 deep learning models. This resulted in 9 overall combinations of face recognition techniques to be compared. To implement a fair comparison of these 9 combinations, first, training and test datasets were gathered; these datasets were made up of complete frontal and well-cropped (NxN sized) face images of 33 persons (mostly Asian), with at least 10 different face images for each person. Then, face images in the training dataset were introduced into deep learning models to extract face vectors from them. Next, these face vectors were stored in a local directory as a reference database (to be used with Euclidean and Cosine distance methods) and were used to train SVM classifier. Subsequently, these face vectors were utilized to classify (recognize) face images (vectors) from the test dataset. As experiments proved, the best face recognition technique amongst 9 combinations was Dlib based face recognition model (with SVM classifier combined) as it showed the highest rate to distinguish people from each other.Although, this research work does not bring novelty to the domain, it took an effort to evaluate/compare well-known deep face models performances on Asian faces (mostly) and choose the best one to utilize as a basis for door access control application.

Published in: 2021 IEEE International Conference on Smart Information Systems and Technologies (SIST)

Date of Conference: 28-30 April 2021

DOI: 10.1109/SIST50301.2021.9465952

Date Added to IEEE Xplore: 29 June 2021

Publisher: IEEE

☰ Contents

I. INTRODUCTION

Face recognition task has been a main focus for a multitude of research work within Computer Vision field in recent decades [22], [23], [24], [25]. The main goal in this task is to detect a face (person) from an image or video frames coming from a video stream, extract face features distinct to that face, and identify/classify that face (person). Starting from 1960s (during 1964-1965, pioneers of automated face recognition [Sign in to Continue Reading](#) [Leon Woody Bledsoe](#), [Helen Chan](#), [Wolf](#), and [Charles Bisson](#) started working on their human face recognition project), there have been developed many algorithms to recognize human faces. These include traditional facial features and landmarks based algorithms [1], [19], [20], [21], [22], skin texture analysis [13] and so on.

Authors	▼
Figures	▼
References	▼
Citations	▼
Keywords	▼
Metrics	▼

More Like This

Face Recognition Based on Principal Component Analysis and Support Vector Machine Algorithms
2021 40th Chinese Control Conference (CCC)
Published: 2021

Building an Extensive Database for Training Predictive Models in Image Classification of Mammography Views and Projections using Support Vector Machines
2023 20th International Conference on Electrical Engineering, Computing Science and Automatic Control (CCE)
Published: 2023

Show More

IEEE Personal Account

CHANGE USERNAME/PASSWORD

Purchase Details

PAYMENT OPTIONS

VIEW PURCHASED DOCUMENTS

Profile Information

COMMUNICATIONS PREFERENCES

PROFESSION AND EDUCATION

TECHNICAL INTERESTS

Need Help?

US & CANADA: +1 800 678 4333

WORLDWIDE: +1 732 981 0060

CONTACT & SUPPORT

Follow

f

@

in

▶

About IEEE Xplore | Contact Us | Help | Accessibility | Terms of Use | Nondiscrimination Policy | IEEE Ethics Reporting | Sitemap | IEEE Privacy Policy

A not-for-profit organization, IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.

© Copyright 2024 IEEE - All rights reserved.

- IEEE Account
- » Change Username/Password

» Update Address
- Purchase Details
- » Payment Options

» Order History

» View Purchased Documents
- Profile Information
- » Communications Preferences

» Profession and Education

» Technical Interests
- Need Help?
- » US & Canada: +1 800 678 4333

» Worldwide: +1 732 981 0060

[» Contact & Support](#)

[About IEEE Xplore](#) | [Contact Us](#) | [Help](#) | [Accessibility](#) | [Terms of Use](#) | [Nondiscrimination Policy](#) | [Sitemap](#) | [Privacy & Opting Out of Cookies](#)

A not-for-profit organization, IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.
© Copyright 2024 IEEE - All rights reserved. Use of this web site signifies your agreement to the terms and conditions.